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Summary

GDG Solution Challenge 2025 Top 105 Global Finalist (64,000+ participants) and 1st-place hackathon winner. Final-year B.Tech (IT) student with proven experience shipping **full-stack web applications** in **React.js, Node.js, and REST APIs**. Built three end-to-end systems spanning blockchain smart contracts, computer vision, and AI-driven content detection — individually and in teams. Seeking Software Engineer / Full Stack Developer roles.

Education

Rungta College of Engineering and Technology — Bhilai, CG
Bachelor of Technology in Information Technology | CGPA: 7.65 / 10 — *June 2022 – May 2026*

Inter Mathurasini Mahavidyalaya Rajauli Nawada — Nawada, Bihar
Class XII (Intermediate) | Percentage: 70% — *Apr 2019 – May 2021*

DAV Public School Daudnagar Aurangabad — Aurangabad, Bihar
Class X | Percentage: 84.6% — *2019*

Technical Skills

Languages: JavaScript (ES6+), Java, Python, C, C++, SQL, HTML, CSS
Frontend: React.js, Hooks, Context API, Redux, Tailwind CSS, SPA Development, Responsive UI
Backend & APIs: Node.js, Express.js, RESTful API Design, JWT Authentication, Solidity, Ethers.js
Databases & Tools: MySQL, MongoDB, Firebase/Firestore, IPFS, Git, Docker, AWS (Foundations)

Projects

Blockchain-Based Certificate Generation & Validation | *React.js, Node.js, Solidity, IPFS, Hardhat* | GitHub

- Built a **React.js (Vite)** frontend with QR-code-based certificate lookup, integrated against a **Node.js/Express** backend issuing certificates as **ERC-721 NFTs** on Ethereum Sepolia Testnet via Hardhat and Ethers.js.
- Designed two core REST API endpoints (`/issueCertificate`, `/verifyCertificate`) integrating **Alchemy RPC** and **Pinata/IPFS** off-chain storage with **JWT authentication**, cutting verification from days to near-instant blockchain lookup.
- Implemented a `issueCertificate()` **Solidity smart contract** storing metadata on IPFS, making credentials cryptographically forgery-proof with no central authority — supporting **mobile-responsive dark/light mode UI**.

AI-Based Intellectual Property Protection System | *Python, React.js, Node.js, YOLOv5, TensorFlow* | GitHub

- Designed a **3-page React dashboard** (Home, VideoAnalysis, DMCARequests) with **Context API** state management, real-time violation notifications, and dark/light mode — enabling non-technical users to review and submit takedowns without backend access.
- Integrated **YouTube Data API** scraper with **OpenCV** frame extraction and **YOLOv5 (ONNX runtime)** object detection, feeding results into an automated `dmca_handler.py` → `dmcaController.js` pipeline stored in **Firestore**.
- Architected a modular **Node.js/Express REST API** (videoRoutes, aiRoutes, dmcaRoutes) with **JWT auth middleware** and centralized error handling, enforcing clean separation of concerns across all service layers.

Mobile-Based Real-Time Floor Plan Detection & 3D Visualization | *MATLAB, OpenCV, AR* | GitHub

- Engineered a **6-stage computer vision pipeline** in MATLAB — from live **DroidCam** mobile feed through grayscale conversion, **Canny edge detection**, **Hough Transform** wall extraction, to pseudo-3D geometric extrusion in a single unified system.
- Implemented **AprilTag-based pose estimation** with calibrated camera parameters (`cameraParams.mat`) to overlay live 3D AR structures on detected floor plans, reducing spatial drift in real-time marker tracking.
- Delivered a **6-mode interactive interface** with camera calibration and multi-stage visualization, producing a deployable low-cost AR prototype for architectural planning on consumer mobile hardware.

Achievements

Top 105 Finalist – Google Developer Group (GDG) Solution Challenge 2025 — Global
Selected from 64,000+ participants across 3,700+ projects worldwide — *2025*

1st Place – Project Competition 2025 (IT Dept., 6th Sem) — RCET Bhilai
Won for AI-driven full-stack IP detection system, judged on production quality and system design — *2025*